

Multipurpose Configurable AUDIO BUFFER For Headphones

PETRE TZV PETROV

This article presents a simple, low-cost stereo headphones buffer using two LM358 operational amplifiers (op-amps). It is used to connect the headphones to line outputs capable of driving loads up to 600-ohm.

The circuit has a high input impedance, low quiescent current and large voltage operating range. Power can be taken either from a USB or an external DC low-noise power supply. The circuit can also be connected to an

MP3/MP4 player or any other audio equipment.

Many audio equipment, including PCs, have line output, some sort of auxiliary audio output or other low-power audio output with signals from around 0.3V rms up to around 1.3V rms (refer Table I). These outputs have sufficient amplitude to drive the headphones but usually do not have sufficient current. Outputs can drive loads having resistances around 600-ohm.

If you connect commonly-available headphones with resistance between 32-ohm and 200-ohm, it may damage these outputs. So, you need a current booster with low-impedance-driving capabilities.

Selection of the op-amp

ICs like LMV358 and LM358 are one of the most inexpensive op-amps available. LMV358 is a low-voltage, low-power op-amp working in 2.7V - 5.5V power supply range. It has very low quiescent current (typically, only 0.410mA). It is available only in SMD

packages. Also, it has relatively low output current but can still be used to drive headphones. This is because many headphones require only 1mW to produce an audible sound.

LM358 is available in both classical DIP/DIL and SMD packages. It provides a lot more output current as compared to LMV358, and is also appropriate for a single power supply. LM358 has the following features:

- Output can go almost to ground with a single positive power supply.
- Output sink and source currents are typically 20mA and 40mA, respectively.
- Quiescent current is typically 3mA.
- It is suitable for battery-operated applications, including 6V, 12V and 24V batteries.
- It has gain product bandwidth of around 1MHz.
- Slew rate is typically around 1V/ μ s.
- Power supply range is from 3V to 32V.

TABLE I
**AUDIO LEVELS (IN VOLTS)
OF SOME LINE OUTPUTS OR
AUXILIARY AUDIO OUTPUTS
IN AUDIO EQUIPMENT**

Rms	V _{peak}	V _{peak-to-peak}
0.316	0.447	0.894
0.7746	1.1	2.19
1.2276	1.74	3.47

Note: 0dBm = 1mW, which is equivalent to 0.7746V rms into 600-ohm load

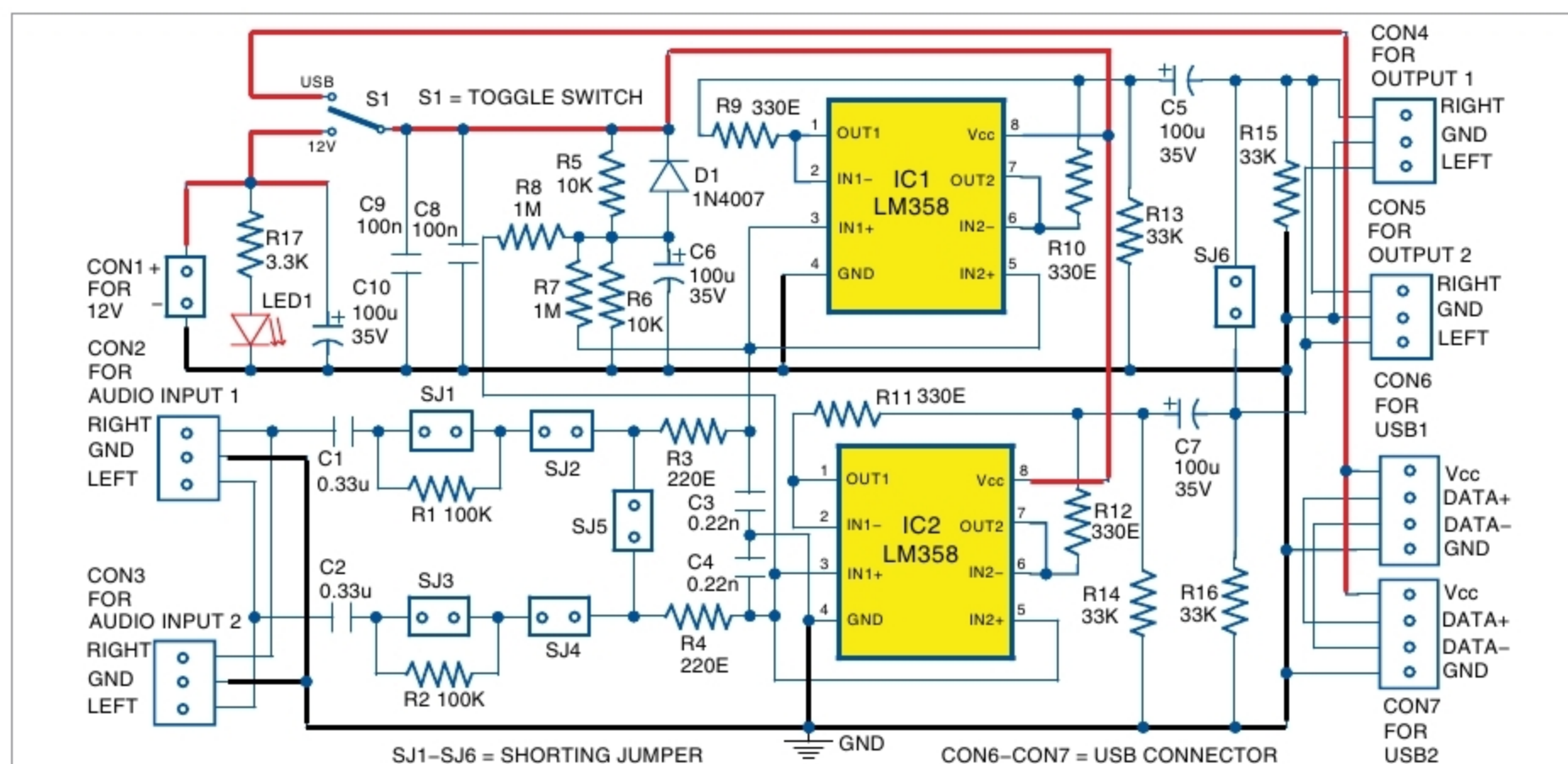


Fig. 1: Circuit diagram of configurable audio buffer for headphones